



eYantra

Engineering a better tomorrow

ERTS Lab
Department of Computer Science and Engineering
Indian Institute of Technology Bombay,
Powai, Mumbai-400 076.



Certificate of Merit

This is to certify that **Sugali Venkata Teja Naik**, a student of Dr. Y C James Yen Government Polytechnic has participated in the **eLSI Robotics Sprint I (2026)**.

He/She is a member of the team having the following team members:

1. Sugali Venkata Teja Naik
2. Shaik Thanveer Basha

The team successfully completed all tasks of the competition and secured rank **10** on the final leader board.

Prof. Shivaram Kalyanakrishnan
Principal Investigator, e-Yantra
Associate Professor
Department of Computer Science and Engineering
Indian Institute of Technology Bombay



eLSI-2026-000652

eLSI Sprint-01

Sprint Description:

The objective of this project is to design and implement an autonomous line-following and object-delivery robot.

The robot begins its operation at a designated start and autonomously navigates a predefined white-line track that includes curves, zigzag sections, and discontinuous (dotted) segments. Along this route, the robot must detect and collect color-coded boxes. Upon successful collection, the robot transitions to a black-line track to the Drop Zones of the boxes. The robot must correctly identify the designated delivery zone for each box.

Participants are expected to design robust control algorithms with PID/RL to ensure reliable line tracking across varying path geometries, accurate color-based crate identification, precise pickup and drop actuation, and efficient task sequencing.

Sprint Learning: PID Control, Reinforcement Learning, Control, Python, Embedded-C, Build-a-Bot.

Certificate Level Matrix

Certificate Level	Certificate Description
Level 1	Certificate of Merit
Level 2	Certificate of Completion
Level 3	Certificate of Participation
Level 4	Letter of Participation

* Level 1 > Level 2 > Level 3 > Level 4 (Level 1 is highest achievement)

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